بسم الله الرحمن الرحيم

أساسيات الحاسب

Computer Fundamentals

Arduino - Line Follower robot

اعداد

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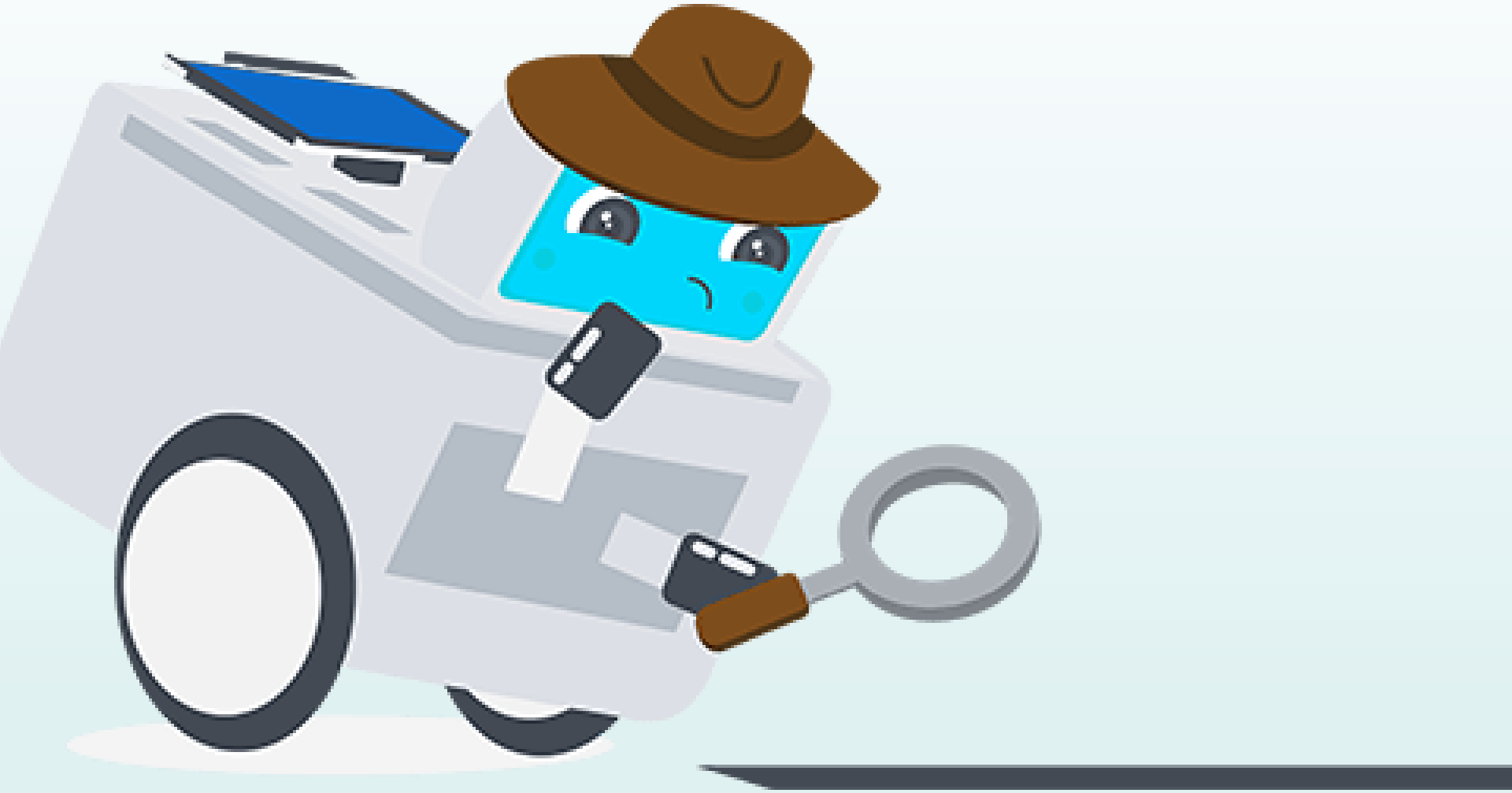

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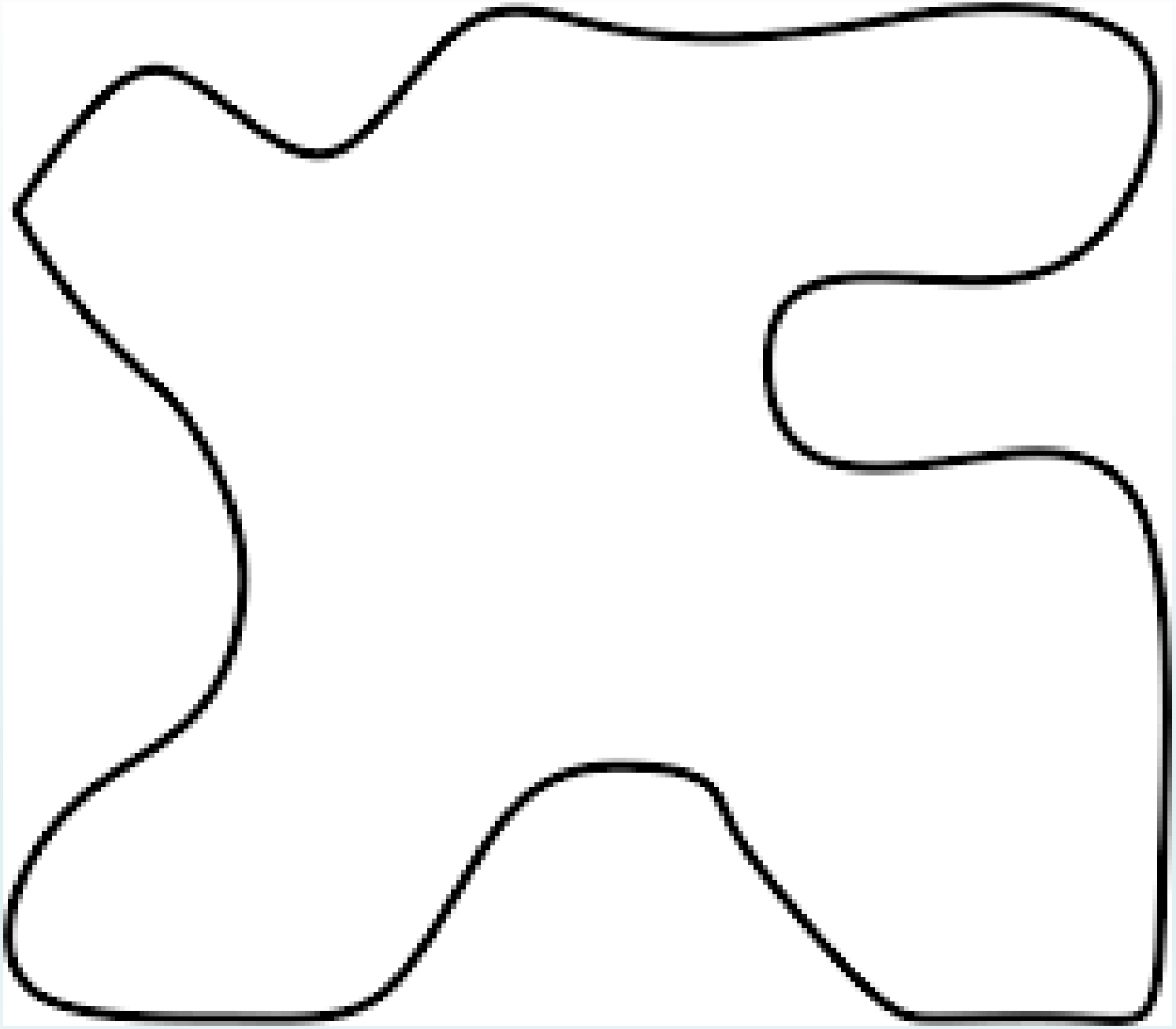
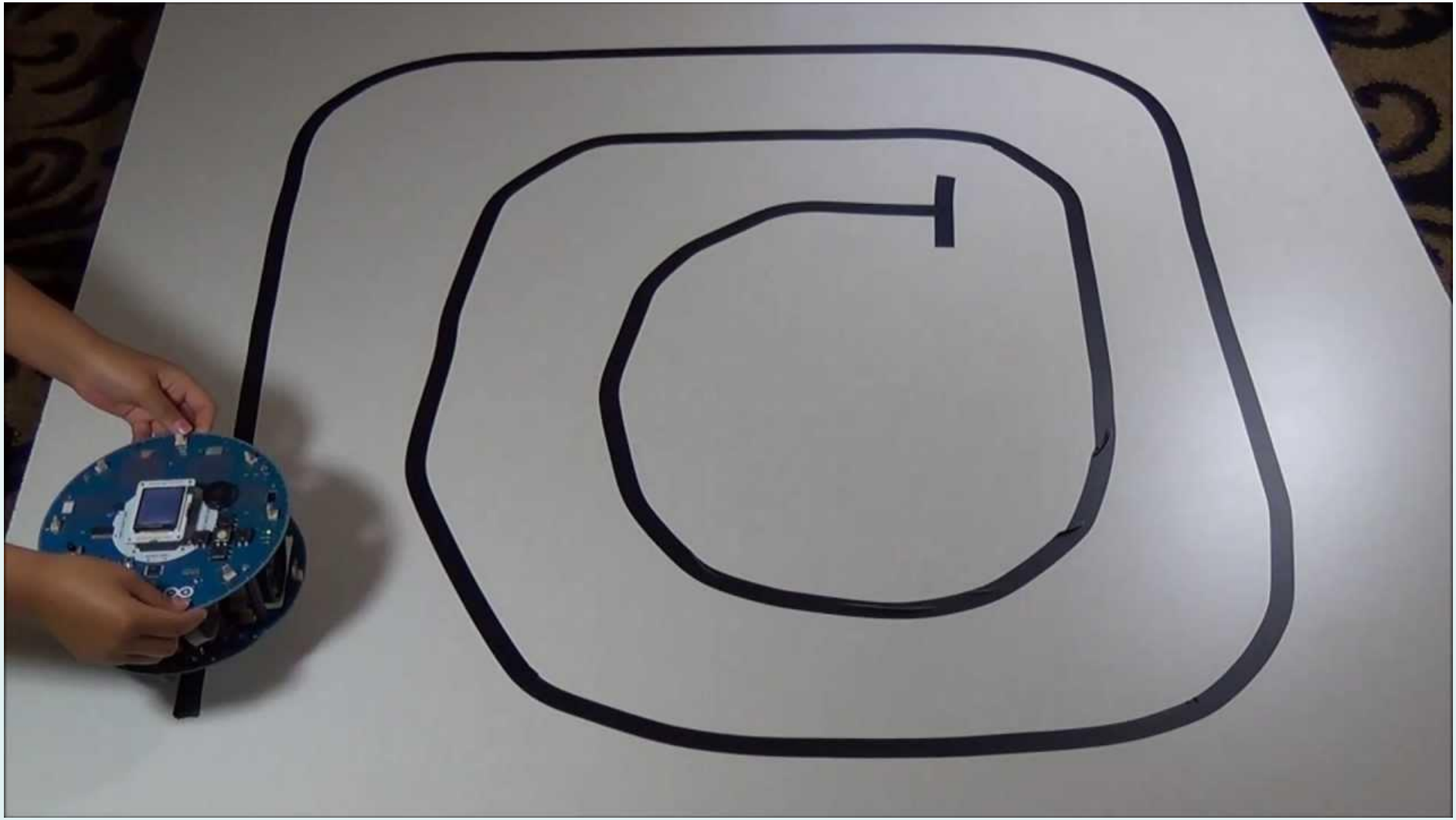
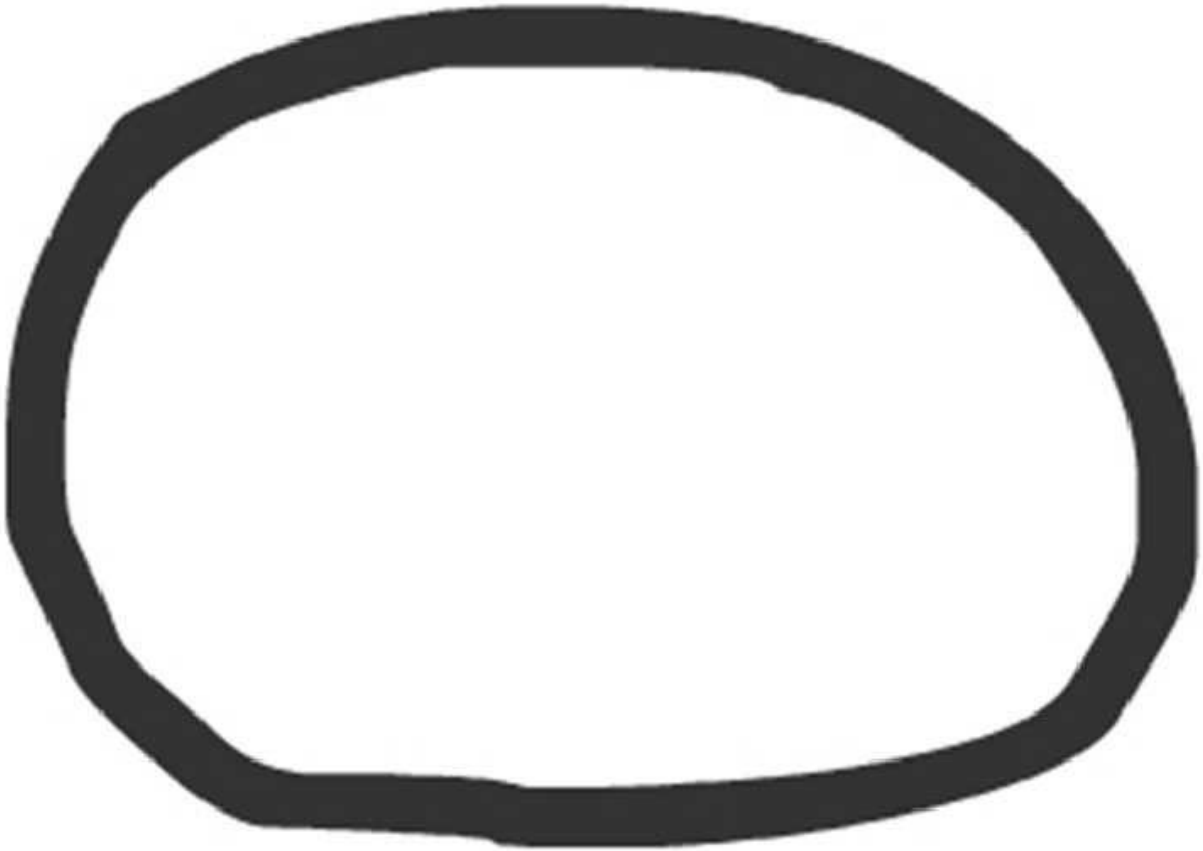
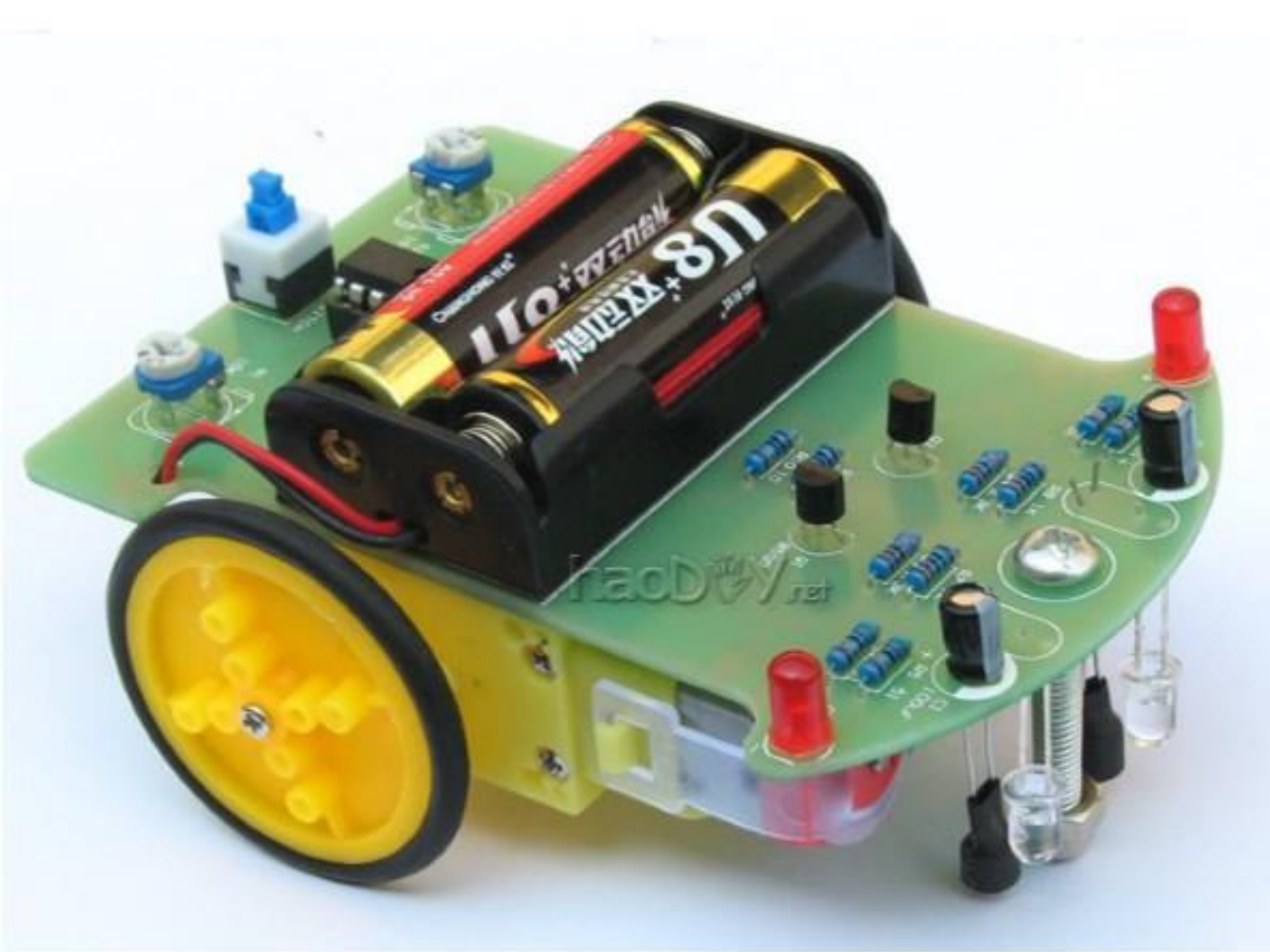


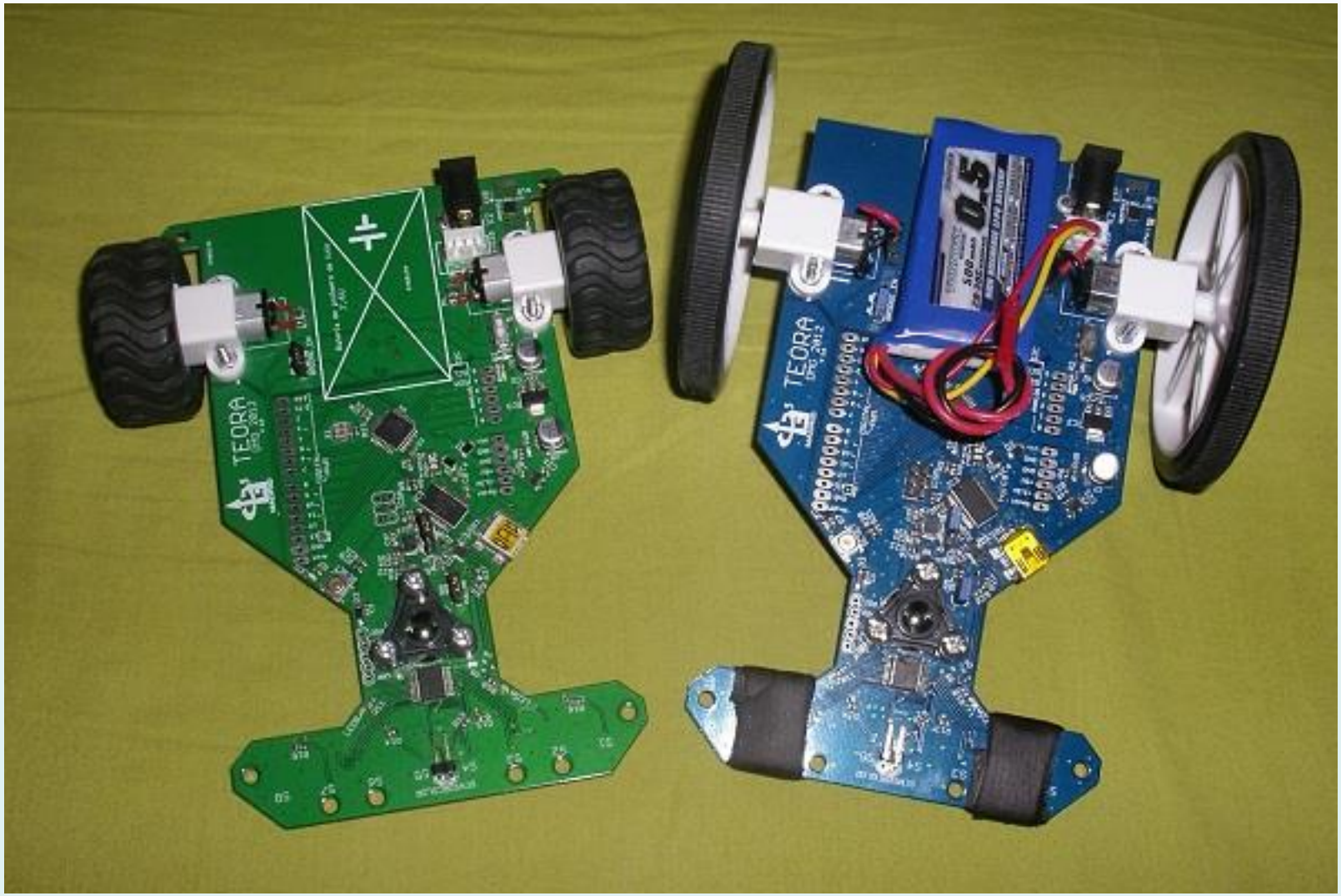






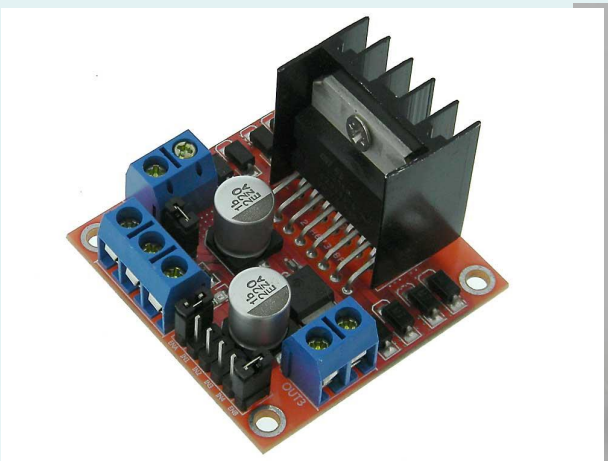
RobotEdu.my

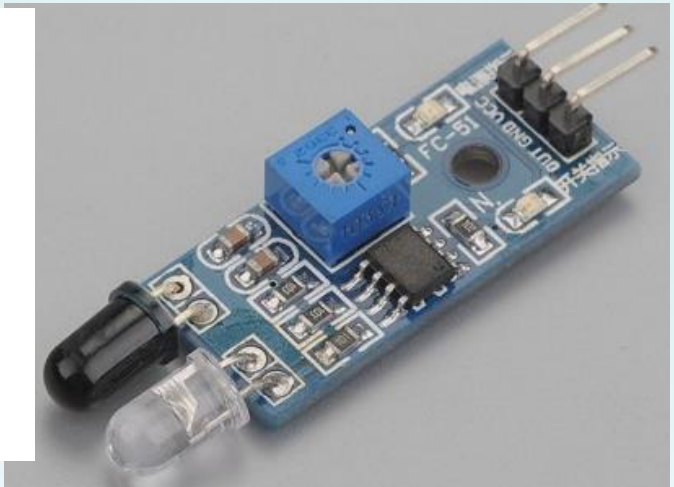
Lowest Price For Better Quality


TEORA
TEORA

KWECOMMERCE

UNO
ARDUINO

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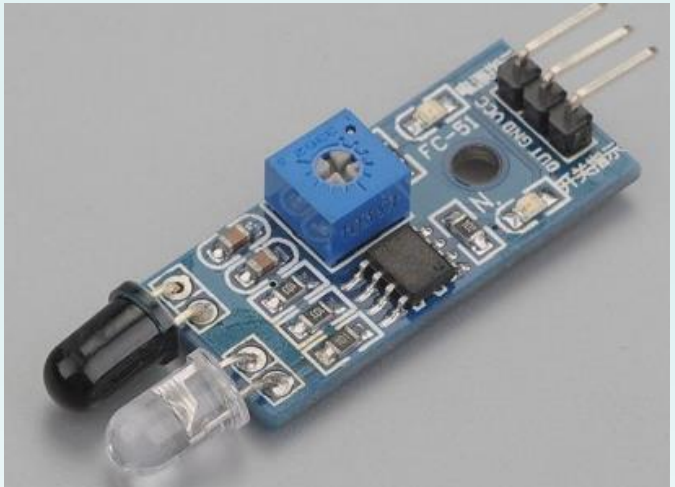
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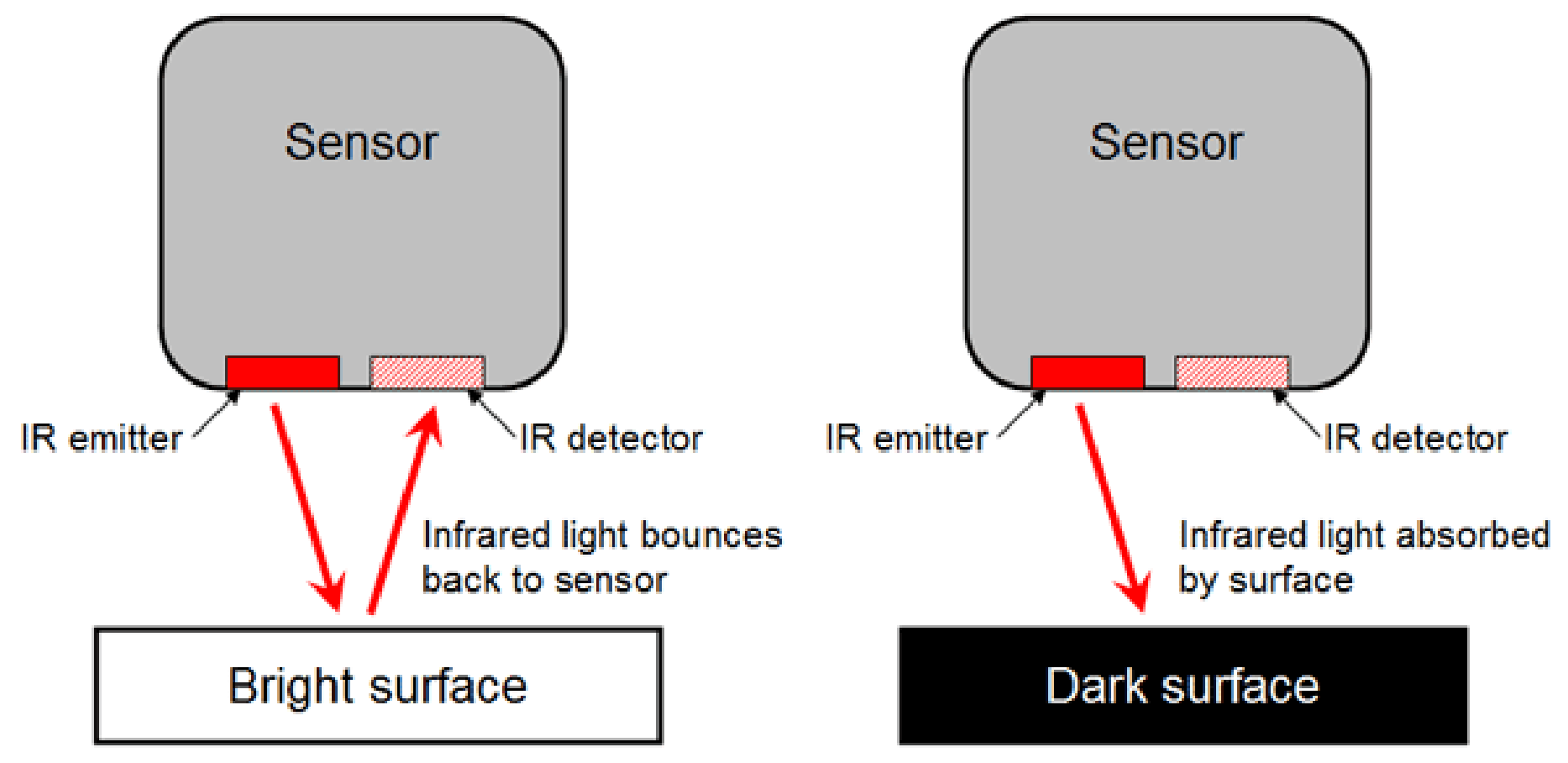
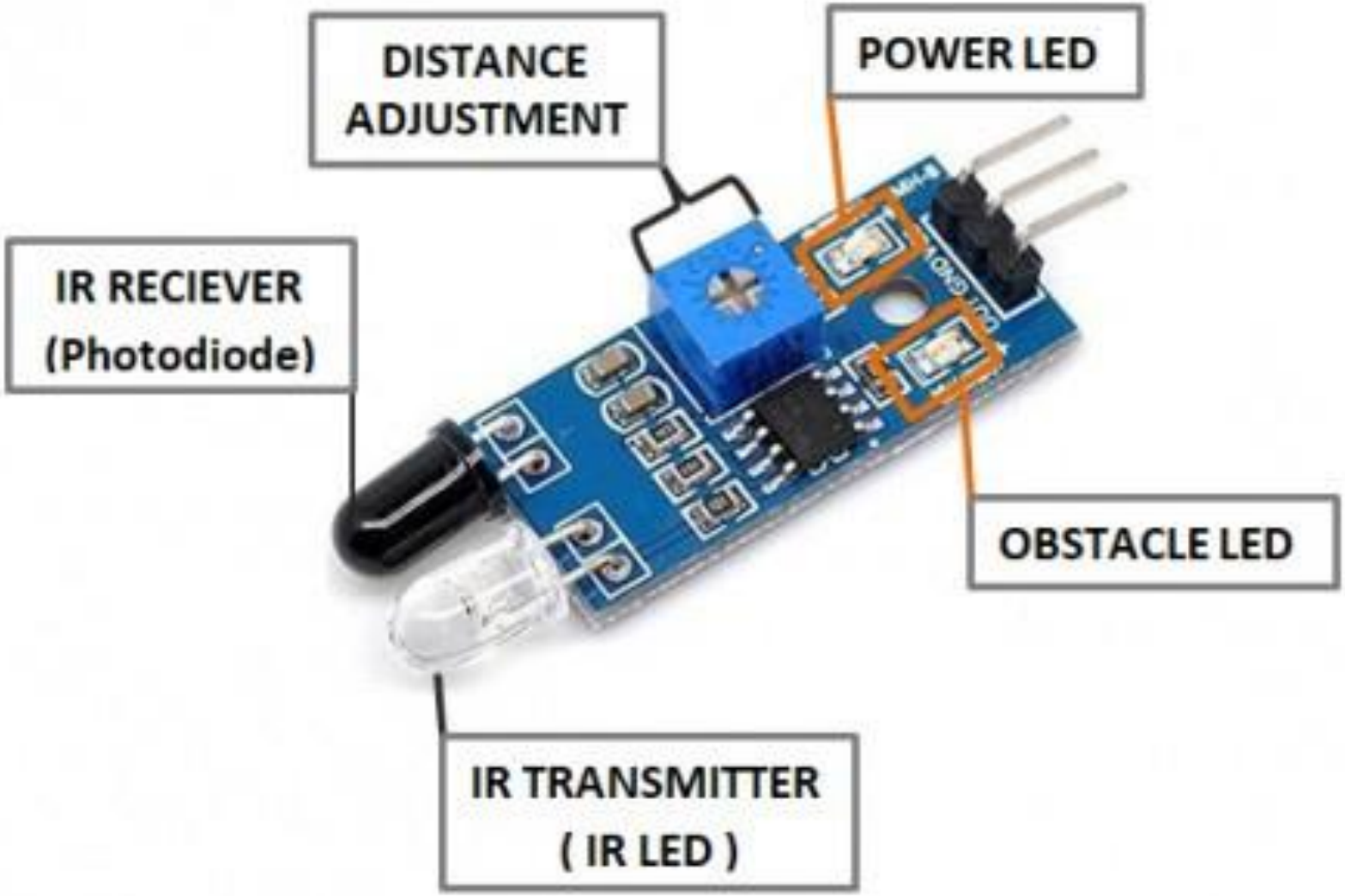
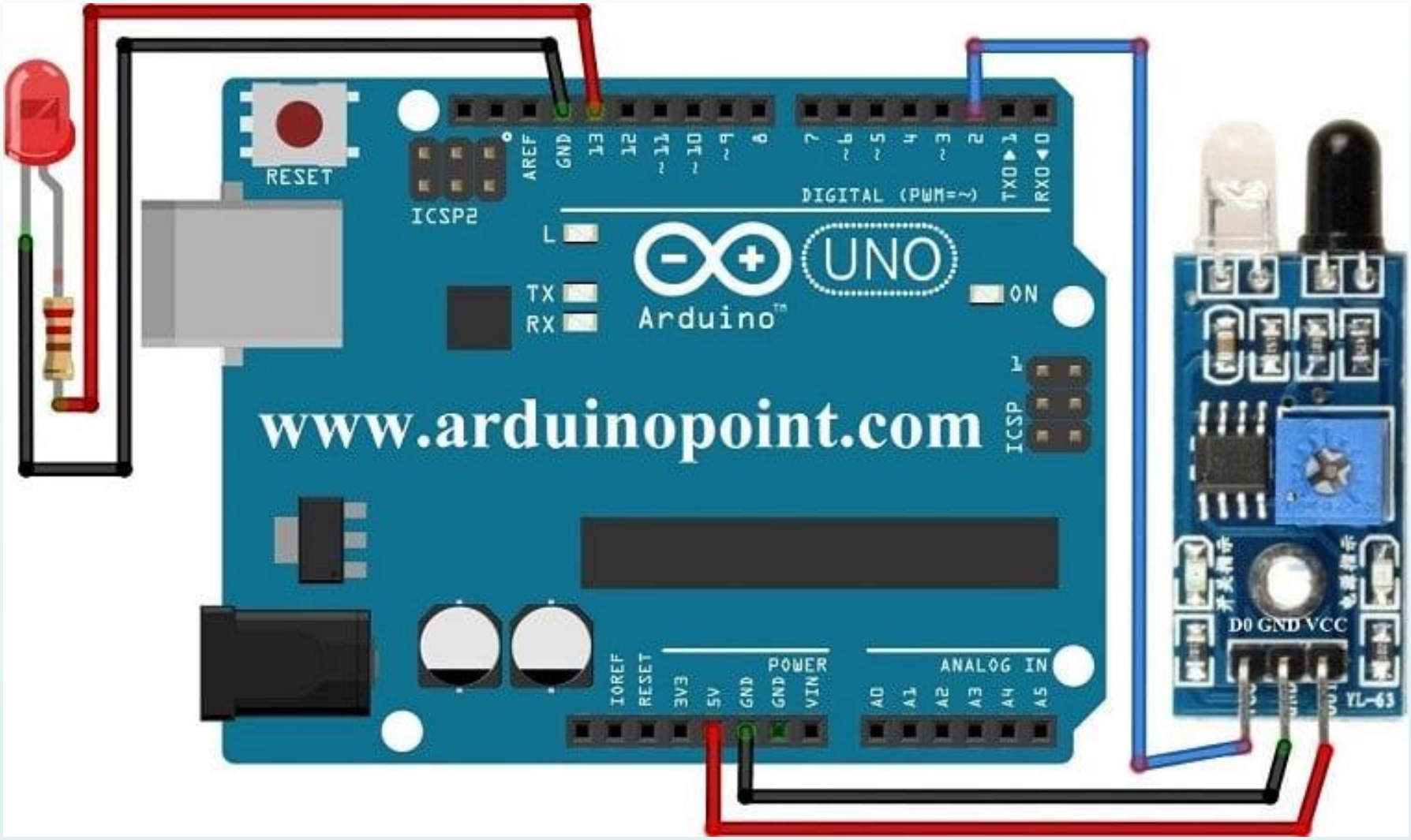

Sensor
IR emitter
IR detector
Infrared light bounces back to sensor
Bright surface
Sensor
IR emitter
IR detector
Infrared light absorbed by surface
Dark surface


DISTANCE ADJUSTMENT
POWER LED
IR RECIEVER (Photodiode)
OBSTACLE LED
IR TRANSMITTER (IR LED)

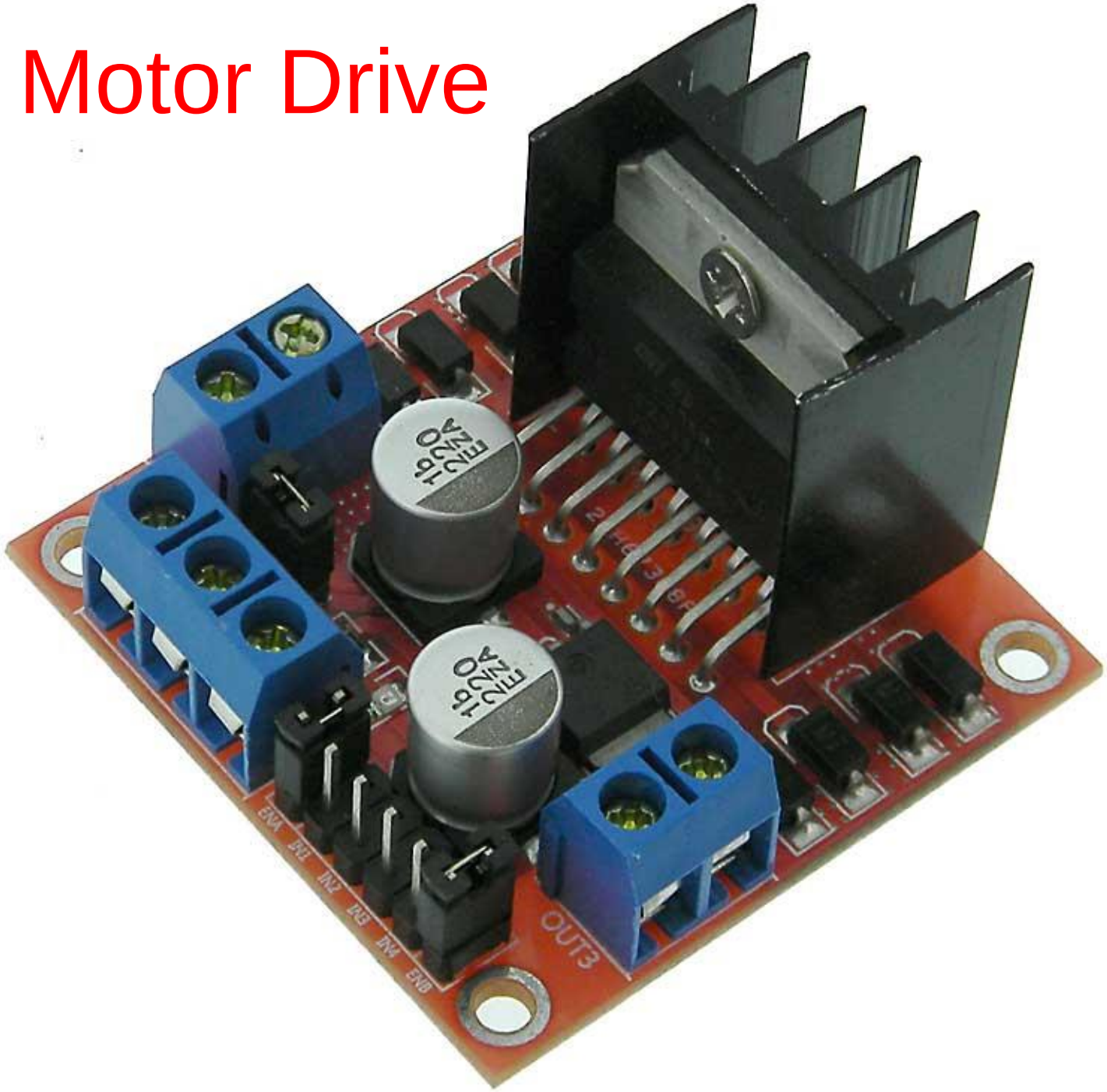
IR Optical Sensor with Arduino

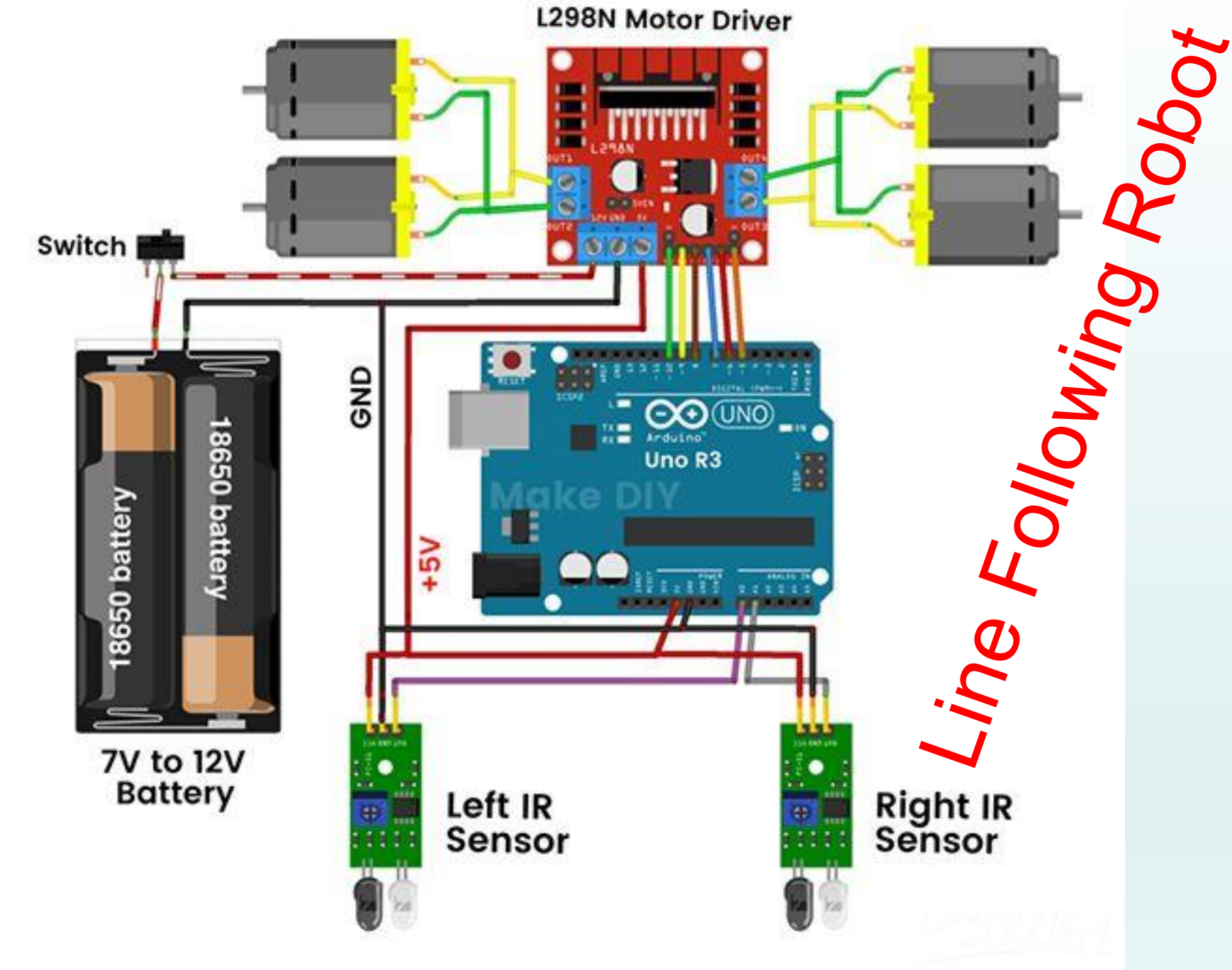


```
int LED = 13; // Use the onboard Uno LED
int isObstaclePin = 2;  // This is our input pin
int isObstacle = HIGH;  // HIGH MEANS NO OBSTACLE

void setup() {
  pinMode(LED, OUTPUT);
  pinMode(isObstaclePin, INPUT);
  Serial.begin(9600);
}
void loop() {
  isObstacle = digitalRead(isObstaclePin);
  if (isObstacle == LOW)
  { Serial.println("OBSTACLE!!, OBSTACLE!!");
    digitalWrite(LED, HIGH);
  }
  else
  { Serial.println("clear");
    digitalWrite(LED, LOW);
  }
  delay(200);
}
```

DC Motor Drive




L298N Motor Driver
Switch
GND
+5V
18650 battery
18650 battery
7V to 12V
Battery
Uno R3
Left IR
Sensor
Right IR
Sensor
Line Following Robot

The Code

```
#define LS 2 // left sensor
#define RS 3 // right sensor
#define LM1 5 // left motor M1a
#define LM2 4 // left motor M2a
#define RM1 7 // right motor M2a
#define RM2 6 // right motor M2b
void setup()
{
pinMode(LS, INPUT);
pinMode(RS, INPUT);
pinMode(LM1, OUTPUT);
pinMode(LM2, OUTPUT);
pinMode(RM1, OUTPUT);
pinMode(RM2, OUTPUT);
pinMode(10,OUTPUT);
pinMode(11,OUTPUT);
analogWrite(10,100);// control the spead of left motor
analogWrite(11,100);// control the speed of right motor
}
```

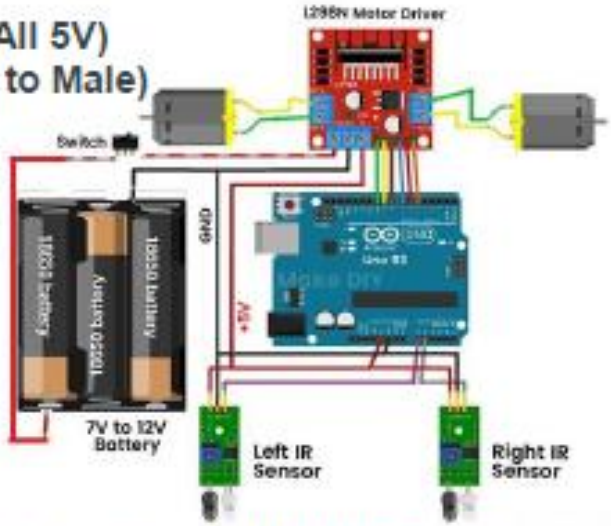
```
void loop()
{// Move Forward on line
if(!digitalRead(LS) && !digitalRead(RS))
{
digitalWrite(LM1, HIGH);
digitalWrite(LM2, LOW);
digitalWrite(RM1, HIGH);
digitalWrite(RM2, LOW);
}
/* turn left by rotationg left motors in
forward and right ones in backward
direction*/
if(digitalRead(LS) && !(digitalRead(RS))) {
digitalWrite(LM1, HIGH);
digitalWrite(LM2, LOW);
digitalWrite(RM1, LOW);
digitalWrite(RM2, HIGH);
}
```

```
/* Turn right by rotating right motors in
forward and left ones in backward
direction*/

if(!(digitalRead(LS)) && digitalRead(RS))
{
digitalWrite(LM1, LOW);
digitalWrite(LM2, HIGH);
digitalWrite(RM1, HIGH);
digitalWrite(RM2, LOW);
}
// Finish line, stop both the motors
if((digitalRead(LS)) && (digitalRead(RS))) {
digitalWrite(LM1, LOW);
digitalWrite(LM2, LOW);
digitalWrite(RM1, LOW);
digitalWrite(RM2, LOW);
}
}
```

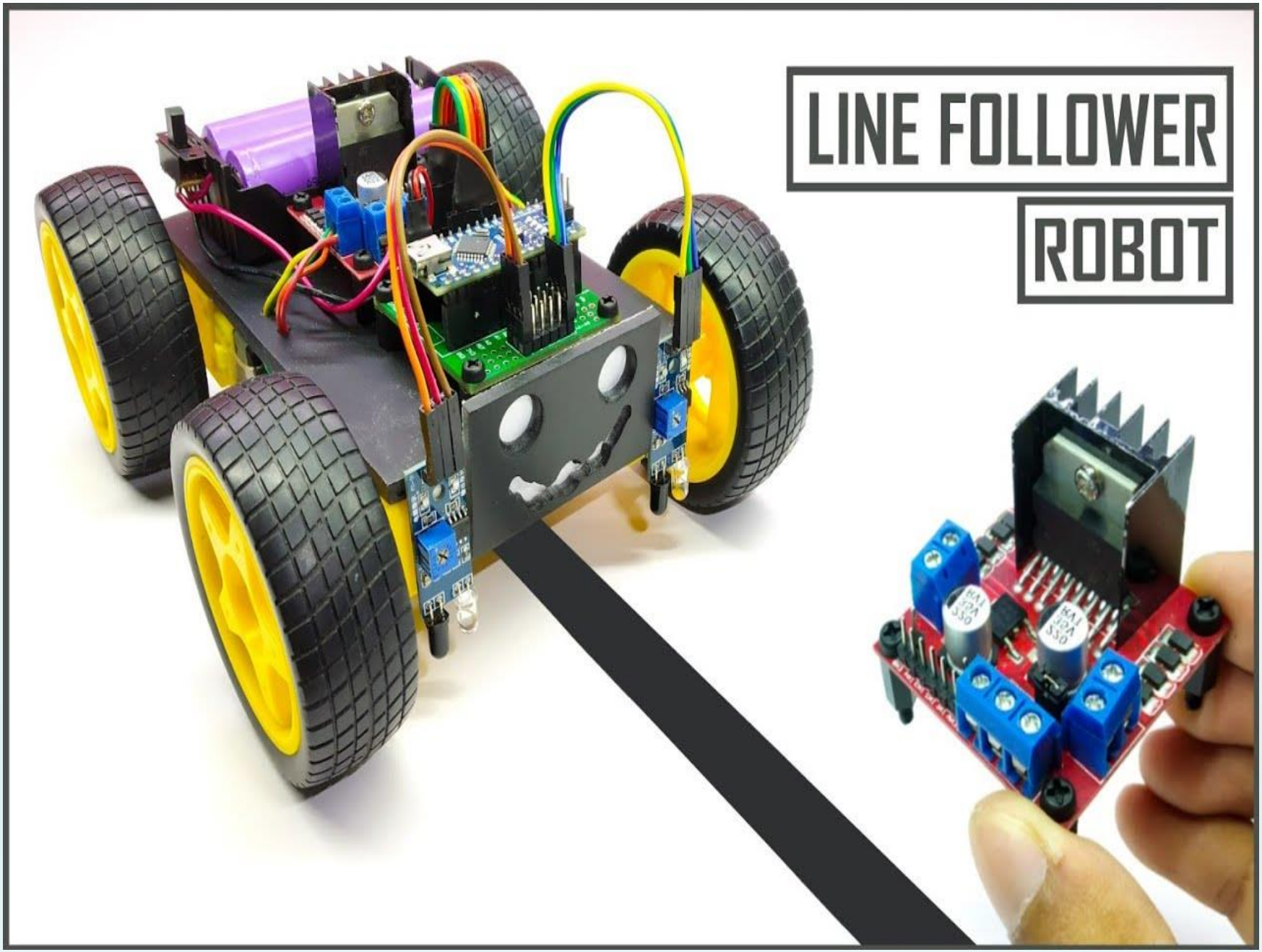
Line follower Robot

- Arduino UNO R3 x1
- USB cable x1
- MINI Breadboard x1
- Red LED x5, Green LED x5, Yellow LED x5 (All 5V)
- Set of wires:(15 Male to Female and 15 Male to Male)
- Buzzer (piezo buzzer) x1
- Resistances 5 x 220ohm, 5 x 1Kohm
- DC motor with wheels x2
- Mini Free wheel x1
- DC motor drive x1
- IR sensor x2
- Mini Switch
- Battery 3.7V x3 and Holder

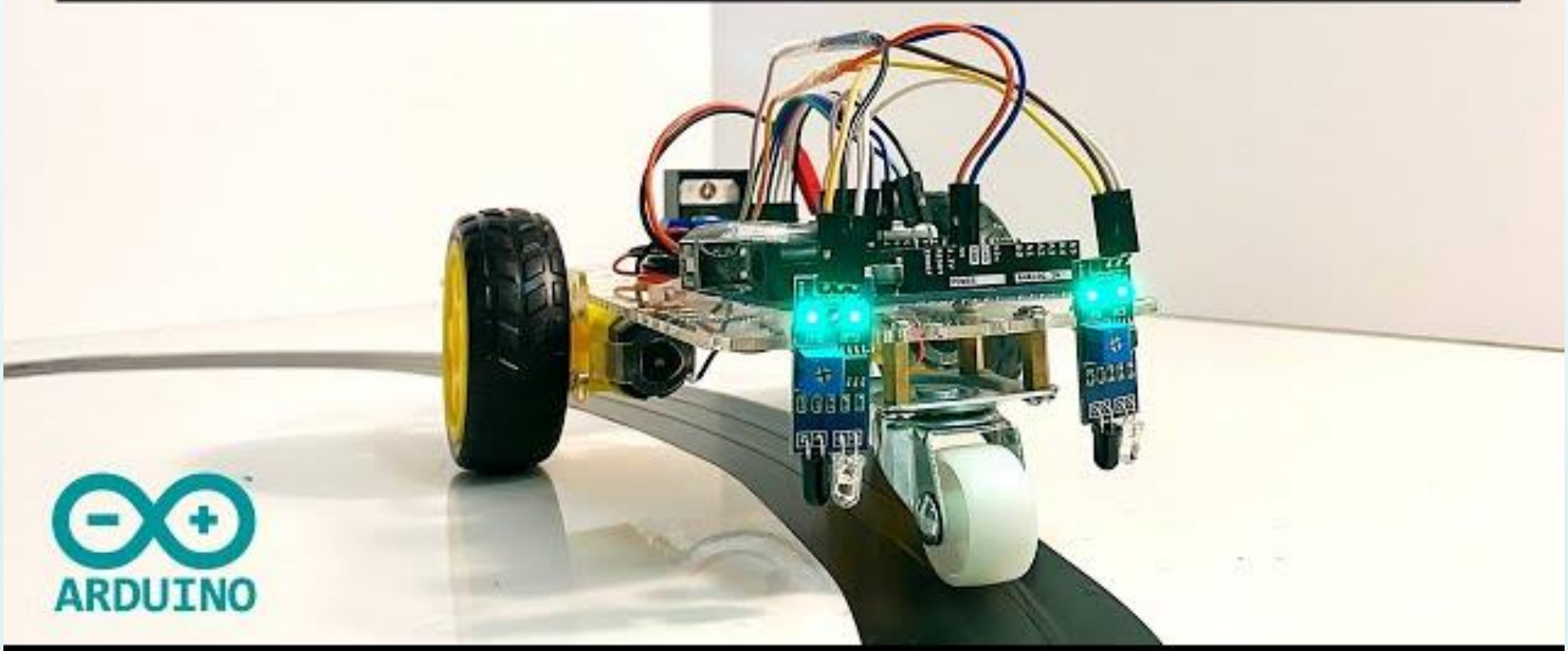


Robot Chassis (Optional)

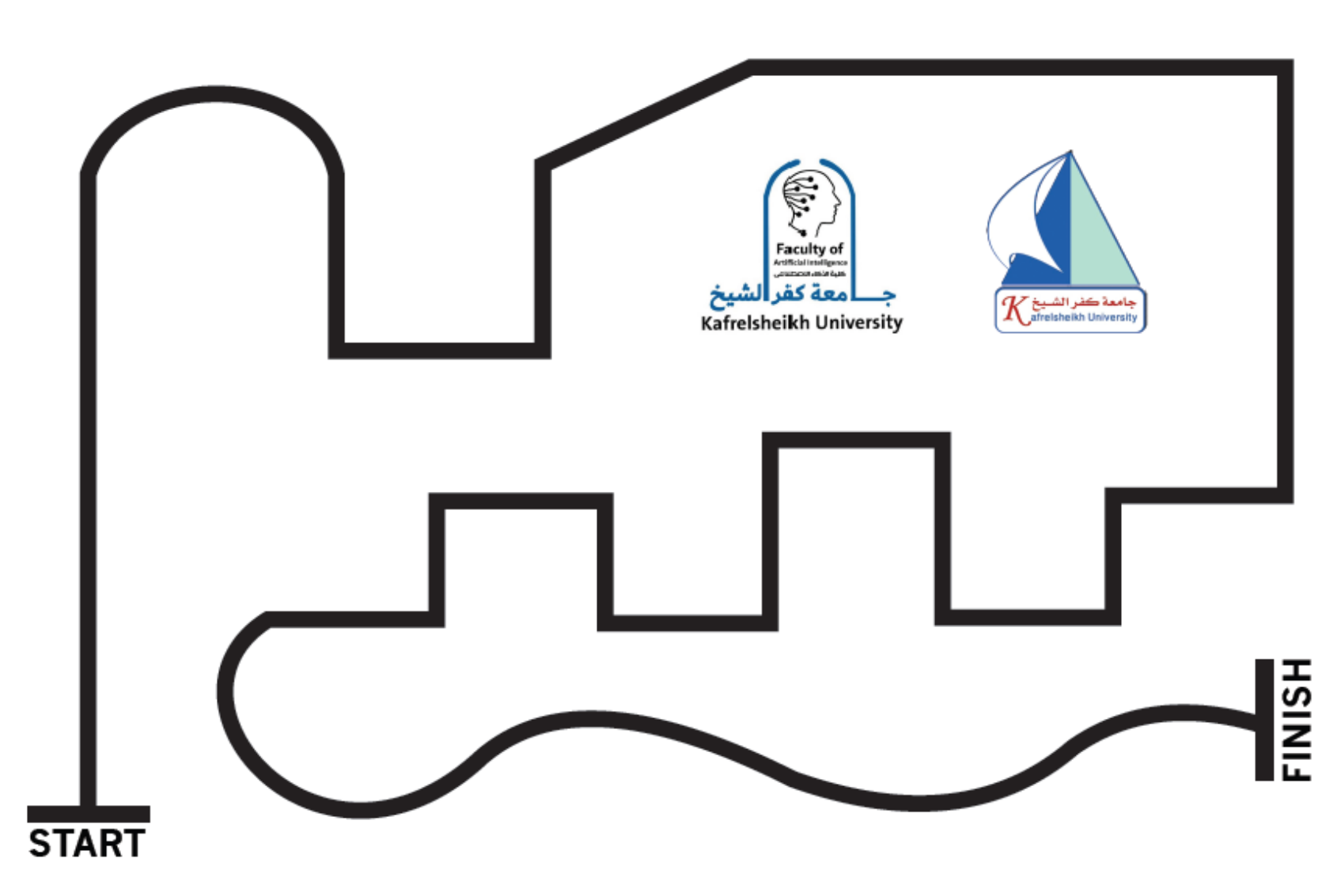
or (جسم الروبوت شاسيه جاهز أو أى قطعة بلاستيك مقوى أو فايبر أو خشب أو كرتون مقوى مقاس 12سم*18سم)


LINE FOLLOWER
ROBOT

LINE FOLLOWER ROBOT



ARDUINO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START
FINISH

Questions ?

Thank you